

Detection of Subclinical Keratoconus Using an Automated Decision Tree Classification

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- **PURPOSE:** To develop a method for automatizing the detection of subclinical keratoconus based on a tree classification.
- **DESIGN:** Retrospective case-control study.
- **METHODS:** SETTING: University Hospital of Bordeaux. PARTICIPANTS: A total of 372 eyes of 197 patients were enrolled: 177 normal eyes of 95 subjects, 47 eyes of 47 patients with forme fruste keratoconus, and 148 eyes of 102 patients with keratoconus. OBSERVATION PROCEDURE: All eyes were imaged with a dual Scheimpflug analyzer. Fifty-five parameters derived from anterior and posterior corneal measurements were analyzed for each eye and a machine learning algorithm, the classification and regression tree, was used to classify the eyes into the 3 above-mentioned conditions. MAIN OUTCOME MEASURES: The performance of the machine learning algorithm for classifying eye conditions was evaluated, and the curvature, elevation, pachymetric, and wavefront parameters were analyzed in each group and compared.
- **RESULTS:** The discriminating rules generated with the automated decision tree classifier allowed for discrimination between normal and keratoconus with 100% sensitivity and 99.5% specificity, and between normal and forme fruste keratoconus with 93.6% sensitivity and 97.2% specificity. The algorithm selected as the most discriminant variables parameters related to posterior surface asymmetry and thickness spatial distribution.
- **CONCLUSION:** The machine learning classifier showed very good performance for discriminating between normal corneas and forme fruste keratoconus and provided a tool that is closer to an automated medical reasoning. This might help in the surgical decision before refractive surgery by providing a good sensitivity in detecting ectasia-susceptible corneas. (*Am J Ophthalmol* 2013;156:237–246. © 2013 by Elsevier Inc. All rights reserved.)

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IDENTIFYING CORNEAS WITH RISK OF DEVELOPING iatrogenic ectasia after laser in situ keratomileusis (LASIK) remains the major concern of the preoperative refractive surgery screening. Although its prevalence is low, estimated between 0.04%¹ and 0.6%,² post-LASIK ectasia is a severe and irreversible complication that compromises the visual prognosis and can lead to corneal transplant.³ Therefore, the great amount of time and energy spent by researchers to develop effective diagnostic tools for identifying patients at risk of such complication remains fully justified.

Identifying subclinical keratoconus (KC) remains one of the most challenging situations faced by the ophthalmologist when considering a refractive surgery procedure. The overlap of values between normal eyes and subclinical KC in most of the parameters analyzed with the topography and tomography does not allow the ophthalmologist to state with certainty to which category the eye belongs. Moreover, the high number and complexity of data provided by the current imaging systems represents a challenge of interpretation for the ophthalmologist. Therefore, decisions are often based on personal experience and subjective recognition of patterns or empiric cutoff values that are not necessarily the same between imaging systems. Recently, some authors have proposed new objective methods for the detection of forme fruste KC based on an artificial intelligence program using a combination of Placido and tomographic corneal indices and reported a high sensitivity. To date, these detection programs have been developed with parameters measured with the Orbscan II system (Bausch & Lomb, Rochester, New York, USA)⁴ and the Sirius system (CSO, Firenze, Italy).⁵

In this study, we describe a new screening program for the detection of forme fruste KC using the GALILEI Dual Scheimpflug Analyzer (Ziemer Ophthalmic Systems AG, Port, Switzerland). The method is based on an automated decision tree classification that helps in discriminating between normal corneas, forme fruste KC eyes, and KC eyes in an objective and quantitative way and without requiring preliminary expertise in interpreting corneal imaging.

METHODS

THIS RETROSPECTIVE CASE-CONTROL STUDY WAS CONDUCTED at the University Hospital of Bordeaux, France, in

the National Reference Center for Keratoconus, and approved by the Institutional Review Board of the Bordeaux University Hospital. The study was conducted in accordance with the tenets of the Declaration of Helsinki and all patients signed specific informed consent to participate in this research.

- **STUDY POPULATION:** A total of 372 eyes of 197 patients were enrolled between September 2011 and May 2012 and imaged with the GALILEI Dual Scheimpflug Analyzer System. Corneas were then classified into 3 groups based on eye conditions: normal (Group 1) included 177 eyes of 95 subjects, forme fruste of KC (Group 2) included 47 eyes of 47 patients, and KC (Group 3) included 148 eyes of 102 patients. Groups were defined as follows:

Group 1. Normal eyes were enrolled among suitable candidates undergoing a screening examination for refractive surgery and among the general population undergoing a routine ophthalmologic examination. All patients had discontinued daily-wear soft contact lens use at least 1 week before evaluation. Eyes were considered normal when no clinical signs of keratoconus and no suggestive topographic or tomographic patterns of suspect keratoconus were found, such as asymmetric bowtie with a skewed radial axes, focal or inferior steepening, central keratometry greater than 47.0 diopters, or corneas thinner than 500 μm . Exclusion criteria for this group were previous ocular surgery, ocular pathology, familial history of keratoconus, and contact lens wearing in the past week.

Group 2. Group 2 was composed of 47 forme fruste of KC, which are defined as the contralateral eyes of clinically evident keratoconus in the fellow eye ($n = 47$).⁶ These eyes had no clinical signs of keratoconus and a normal topographic aspect with no asymmetric bowtie and no focal or inferior steepening pattern. This condition is also known in the literature as “subclinical keratoconus” since it has already been reported that approximately 50% of clinically normal fellow eyes of patients with a unilateral keratoconus progressed to keratoconus within 16 years, with a greater risk during the first 6 years of onset.⁷

Group 3. Eyes with keratoconus were enrolled among patients that were referred to the National Reference Center for Keratoconus for a regular control visit for moderate to advanced keratoconus. Diagnosis of clinical keratoconus was previously defined and includes a combination of findings characteristics of keratoconus^{8,9}: corneal topography with asymmetric bowtie pattern or localized steepening, irregular cornea determined by distortion of the retinoscopic or ophthalmoscopic red reflex, and at least 1 of the following slit-lamp findings: stromal thinning, Fleischer ring greater than 2 mm arc, Vogt striae, and corneal scarring consistent with keratoconus. Eyes that wear contact lenses and eyes that have already

undergone a specific treatment for keratoconus, such as collagen cross-linking, intracorneal rings, or keratoplasty, as well as marginal pellucid degeneration, were excluded from the study.

- **DUAL SCHEIMPFLUG ANALYZER SYSTEM AND PROCEDURE:** Measurements were performed with the GALILEI system (software version 5.2.1) according to the manufacturer's guidelines, and all the measurements were taken by the same experienced examiner (D.S.).

The GALILEI is a rotating Scheimpflug tomography-based device combining dual-channel Scheimpflug cameras and a Placido disk. The rotational scanning system acquires between 15 and 60 Scheimpflug images per scan and 2 Placido top view images at 90 degrees apart, as the cameras rotate around the central axis. Placido and Scheimpflug data are acquired simultaneously, and then a motion correction algorithm is applied to the combined data set. This correction compensates for a patient's eye motion during scanning by a tracker that locates and tracks a patch on the iris, matching its location on every scan. Height data from the Scheimpflug images and slope data, converted into height data from the Placido, are merged to provide a surface fitted to the anterior corneal data, whereas posterior corneal surface data are measured using edge detection in images provided by the dual Scheimpflug system. Simultaneously, the system allows for a corneal aberration analysis separately from the aberrations of the lens and displays the total higher-order corneal wavefront aberrations calculated from the front and back surface. Both the displayed wavefront maps and the root mean square indices are recalculated recentered on the pupil center over a 6.0-mm optical zone. Individual Zernike coefficients for terms from second to sixth order are displayed in micrometers as well as in diopters.

Only measurements that satisfy the minimum quality required by the system were included in this study. Good quality of the measurement was reached when the scan registered as “OK” on the Instrument Quality Checking display. The overall scan quality takes into account the following 4 factors: quality of the Placido images, Scheimpflug images, motion compensation, and motion distance, which appear with respective percentage thresholds to reach for the measurement to be considered of sufficient quality.

- **AUTOMATED MACHINE LEARNING CLASSIFIER: THE TREE CLASSIFICATION METHOD:** *Description.* In our study, we used an automated classification method to discriminate between normal, forme fruste KC, and KC eyes in an objective and quantitative way. The classification and regression tree is a flexible nonparametric data analysis tool that has been adopted for data mining.¹⁰ It is generated by a machine learning algorithm (RPART software, R package version 3.1-50) using a sample of “training” data with a known class assignment (normal, forme fruste KC, or KC, in our case).

Tree interpretation rule **Group A versus Group B**

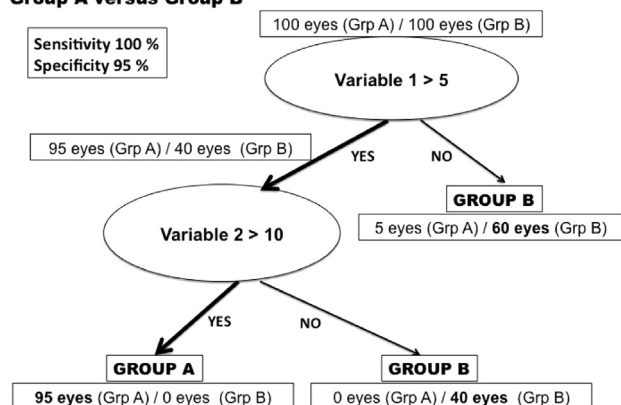


FIGURE 1. Model of discrimination rule generated using an automated decision tree classification for discriminating 2 groups, A and B. The primary split is based on the discriminating variable 1 with a cutoff value of 5. The condition expressed by this first node and that is selected for discriminating between the 2 groups is “variable 1 greater than 5.” The first separation shows that 95 eyes in Group A and 40 eyes in Group B met this criterion, whereas 5 eyes in Group A and 60 eyes in Group B did not. Subsequently, 95 eyes in Group A and 0 in Group B met the condition expressed by the second split (variable 2 greater than 10), whereas 40 eyes in Group B and 0 in Group A did not meet this condition. Therefore, an eye should be classified as belonging to Group A when its variable 1 is larger than 5 and its variable 2 is larger than 10, whereas an eye should be classified as belonging to Group B if its variable 1 is smaller than 5 but its variable 2 smaller than 10. In this particular decision tree, 100 eyes with condition B have been correctly classified (100% sensitivity; no false negative), whereas 95 eyes with condition A were correctly classified and 5 misclassified (95% specificity; 5% of false positive).

The decision tree consists of successive nodes that are defined by a discriminant variable and its cutoff value, branches that successively split the sample into 2 mutual subgroups, and leaves that correspond to the final decision of class assignment (Figure 1). From all measured parameters, the algorithm automatically selects the variable and a cutoff value for that variable that best discriminates between the given groups.

A final decision tree that is too large, including too many nodes (large tree), risks overfitting the training data and thus poorly generalizing to new samples. Therefore, a pruning process is applied to the decision tree (pruned tree) in order to improve the predictive accuracy by omitting parts that provide little additional power and are considered as “noise.”

Interpretation rules. One of the major advantages of the tree classification is its ease of interpretation and of understanding. Moreover, the classification and regression tree has the advantage to fit the medical reasoning by building

its discriminating rule based on a logical sequence of conditional statements “if, then.” Trees are represented graphically, with the root node, which represents the undivided data, at the top, and the branches and leaves beneath. Nodes represent a particular attribute of the data, which correspond to the most discriminant parameter automatically selected by the algorithm, with a specific cutoff value that best discriminates the groups. At each given split, eyes that meet the condition expressed by the parameter (ie, variable <4) are separated from eyes that do not meet this condition. All paths from the root node to the leaf node proceed by way of conjunction, or “and.” That way, final eye conditions assigned at the bottom of the tree will be characterized by a succession of variables with particular cutoff values that will be chosen for discriminating between eyes from different conditions, normal, forme fruste KC, and KC. An artificial example of a decision tree for classifying subjects (healthy vs sick) is represented in Figure 1.

Validation method. The goodness of fit of the tree as a classification tool is characterized in terms of its misclassification rate. These estimated misclassification rates are underestimated because they are evaluated on the basis of the data that produced the classification rules. Thus, in the algorithm that produces the tree it is common to apply a 10-fold cross-validation method.¹¹ This method is applied by randomly splitting the sample into 10 approximately equal-size subsamples. Trees are then produced 10 times. Each time the tree is constructed based on data in which 1 of the subsamples was left out. The omitted part is subsequently used to evaluate the “prediction error” when we apply on these omitted data the classification rule of the tree that was constructed without that omitted part. The final tree is selected to be the one with the smallest cross-validation estimate of error.

The tree classification was built by choosing automatically the most discriminant parameters among 55 variables measured by the system and submitted to the software. All the analyzed parameters are listed in Table 1.

• **ANALYZED PARAMETERS AND DESCRIPTION:** All patients had a detailed preoperative ophthalmic evaluation including uncorrected visual acuity, best spectacle-corrected visual acuity using Early Treatment Diabetic Retinopathy Study (ETDRS) charts, manifest refraction, slit-lamp evaluation, applanation tonometry, and fundus examination.

Both eyes were imaged by the GALILEI Analyzer and 55 parameters were extracted. Briefly, the analyzed parameters can be described as follows:

- **Curvature-based parameters.**
- Mean keratometry (MeanK) is the average keratometry calculated over the analyzed area, central (0-4 mm), paracentral (4-7 mm), or peripheral (7-10 mm) cornea.

TABLE 1. List of Parameters Measured by the GALILEI Analyzer System and Submitted to the Automated Decision Tree Classifier for Discriminating Normal Corneas From Keratoconus and Forme Fruste Keratoconus

Curvature-Derived Parameters	Elevation-Derived Parameters	Wavefront-Derived Parameters	Pachymetric-Derived Parameters	Biometric-Derived Parameters	Other Parameter
Anterior data: Kmax Axial SimKs, Kf, Cyl Axial MeanK (0-4 mm) Tang MeanK (4-7 mm) Tang MeanK (7-10 mm) Eccentricity (ϵ^2) I-S value	Anterior data: With BFTA: Max elevation TP Max elevation Kmax Max elevation With BFS: Max elevation TP Max elevation Kmax Max elevation AAI	RMS total corneal HOA RMS corneal SA RMS corneal vertical coma RMS corneal horizontal coma RMS total corneal coma Corneal power Total corneal power (0-4 mm) Total corneal power (4-7 mm) Total corneal power (7-10 mm) Central CP ratio (0-4 mm) Paracentral CP ratio (4-7 mm) Peripheral CP ratio (7-10 mm)	Thinnest point Corneal volume	Anterior chamber depth Anterior chamber volume	Age
Posterior data: Axial Ks, Kf, Cyl Axial MeanK (0-4 mm) Tang MeanK (4-7 mm) Tang MeanK (7-10 mm) Eccentricity (ϵ^2)	Posterior data: With BFTA: Max elevation TP Max elevation Kmax Max elevation AAI				
Keratoconus indices: DSI, SAI, OSI, SRI, CSI, IAI, ACP, AA, SDP	Max elevation With BFS: Max elevation TP Max elevation Kmax Max elevation AAI				

AA = area analyzed; ACP = average central power; BFS = best fit sphere; BFTA = best-fit toric aspheric sphere; CP = corneal power; CSI = central/surround index; Cyl = cylinder (diopters); DSI = differential sector index; HOA = higher-order aberrations; IAI = irregular astigmatism index; I-S = inferior-superior; ISPS = I-S posterior score; AAI = asphericity asymmetry index; Kf = flattest K; Ks = steepest K; MeanK = average keratometry (diopters); OSI = opposite sector index; SA = spherical aberrations; SAI = surface asymmetry index; SDP = standard deviation power; SRI = surface regularity index; TP = thinnest point.

These values are derived from the axial curvature maps for the central cornea and from the tangential curvature maps for the paracentral and peripheral cornea.

- Kmax has been directly recorded from the curvature map and represents the maximal keratometric value.
- Eccentricity (ϵ^2) is 1 of the 4 parameters by which the shape of a conic section can be described. Q (asphericity), P value, and E (corneal shape factor) are the others. These terms are mathematically related by the following equation: $\epsilon^2 = E = 1 - P = -Q$. It is calculated within a central diameter of 8 mm averaged over all meridians of the anterior corneal surface. A positive value refers to a prolate shape of the corneal surface, whereas a negative value refers to an oblate shape.
- The I-S value, which has been described by Rabinowitz and McDonnell as the amount of steepening of the inferior cornea compared with that of the superior cornea.¹²
- The opposite sector index (OSI), which has been described by Maeda and associates and represents the greatest difference in average area-corrected power between opposite 45-degree sectors.¹³
- Posterior curvature data are basically the same as for the anterior surface; the only difference is that the keratometry values (Ks, Kf) are not simulated since they are

calculated with the real indices of refraction of the cornea (1.376) and the aqueous humor (1.336).

Elevation-based parameters.

- Elevation values were measured with 2 different reference bodies over a fixed 8-mm calculation zone: the best-fit sphere in float mode and the best-fit toric and aspheric body. Values were recorded in both anterior and posterior surfaces over 3 locations by manually guiding the cursor over the anterior and posterior elevation maps: highest elevation value within the 8-mm-diameter zone, elevation value at the thinnest point, and elevation value at the Kmax location.
- The quantification of asymmetry of asphericity of a corneal surface, the asphericity asymmetry index (AAI), has been proposed by Arce (Arce C. Electronic Poster, ASCRS Congress. April 9-14, 2010; Boston, Massachusetts). It is calculated over the best-fit toric and aspheric reference surface map display as the absolute value of the difference between the maximum negative elevation value and maximum positive elevation value within the central 6-mm-diameter data zone. This index has been recorded for the anterior and posterior surfaces.

Corneal wavefront-based parameters. Total corneal higher-order aberration root mean square from the third to the sixth order as well as the root mean square spherical aberration $Z(4,0)$, root mean square vertical $Z(3,-1)$ and horizontal $Z(3,1)$ coma, and root mean square total coma through a 6-mm pupil size were recorded from the wavefront maps, displayed in micrometers.

Enantiomorphism was neutralized by inverting the sign of the mirror-symmetric coefficients of the left eyes as shown in the following equations:

$$\text{For all } C_n^m \text{ if } n \text{ is even and } m < 0 : C_n^m = -(C_n^m)$$

$$\text{For all } C_n^m \text{ if } n \text{ is odd and } m > 0 : C_n^m = -(C_n^m)$$

Corneal power-based parameters. The average total corneal power over 3 different corneal zones, respectively, central (0-4 mm), paracentral (4-7 mm), and peripheral (7-10 mm), was recorded. The total corneal power is the actual power of the cornea including both the anterior and posterior surfaces. The total corneal power and map are calculated using ray tracing by tracking the path of incident rays of light through the 3-dimensional cornea. In addition, the posterior-anterior corneal power ratio over the 3 different analyzed corneal zones was calculated as the ratios of the anterior value to the posterior value.

Pachymetric- and biometric-based parameters. The thinnest point of the cornea, as well as the corneal and anterior chamber volume calculated over a diameter of 8.0 mm, was recorded. Anterior chamber depth was calculated as the distance between the crystalline lens and the posterior cornea, measured along normal to the line between the outer iris endpoints.

- **STATISTICAL ANALYSES:** Two pairs of large and pruned trees were generated based on the different discriminating rules: the rule for discriminating between normal eyes and KC and the rule for discriminating between normal eyes and forme fruste KC. RPART software (R package version 3.1-50) was used to generate the classifications trees. Large trees are descriptive of our studied population and not necessarily applicable with similar performance to new samples, since they have been developed on the basis of our training data. Pruned trees (smaller trees with fewer splits) are generated after applying a pruning process that reduces the size of the decision tree by removing the lower sections of the tree that have a relatively small contribution to the discriminating power. Therefore, pruned trees are considered as “suboptimal trees” since they provide classification rules that will represent a better global solution that will not represent a locally optimal solution that applies only to our sample data.

Part of the statistical analyses concerned the question of dependence between observations. Since for most patients

the data included both eyes, it seemed plausible that there will be an intraclass correlation within patients, so that the whole database cannot be considered as independent observations. Therefore, we initially used the mixed model rather than ANOVA for comparing the groups. The mixed model enabled us to take account of the dependence among observations. However, results of this analysis indicated that there was no dependence, so that the independence assumption was valid within the studied sample.

RESULTS

- **CLASSIFICATION AND REGRESSION TREE DISCRIMINATING RULES: Normal vs keratoconus.** Large and pruned trees for discriminating normal from keratoconic eyes are illustrated in Figure 2. The discriminating rules adopted used, respectively, 3 and 2 variables for the large and pruned trees, and enabled discrimination of normal from KC with 100% sensitivity and 99.5% specificity with the large tree and 99.3% sensitivity and 99.5% specificity with the pruned tree. The most discriminant variable selected by the software for the first split of the decision tree is related to the posterior asphericity asymmetry, the asphericity asymmetry index, with a cutoff value of 34.5 μm .

Normal vs forme fruste keratoconus. Large and pruned trees for discriminating normal eyes from forme fruste KC are illustrated in Figure 3. Respectively, 6 and 2 variables were used for building the discriminating rules for the large and pruned trees. The large tree discriminating rule enabled discrimination of normal eyes from forme fruste KC with 93.6% sensitivity and 97.2% specificity, whereas the pruned tree discriminating rule achieved 90% sensitivity and 86% specificity. The most discriminant variable selected by the software for the first split of the decision tree was also related to the posterior asphericity asymmetry, the asphericity asymmetry index, with a cutoff value of 21.5 μm , whereas the corneal volume, with a cutoff value of 30.8 mm^3 , was selected as the second most discriminant variable for differentiating normal eyes from forme fruste KC (Figure 4).

The performances of the classification trees are summarized in Table 2 and means of the various parameters selected in the decision trees are summarized in Table 3.

DISCUSSION

THE PRELIMINARY QUESTION THAT HAS TO BE ADDRESSED when developing a new method for the detection of subclinical KC is who should we target and label to be at risk of post-LASIK ectasia, independent of surgical factors such as flap thickness, ablation depth, or residual stromal bed. Several terms have been put forward for describing this

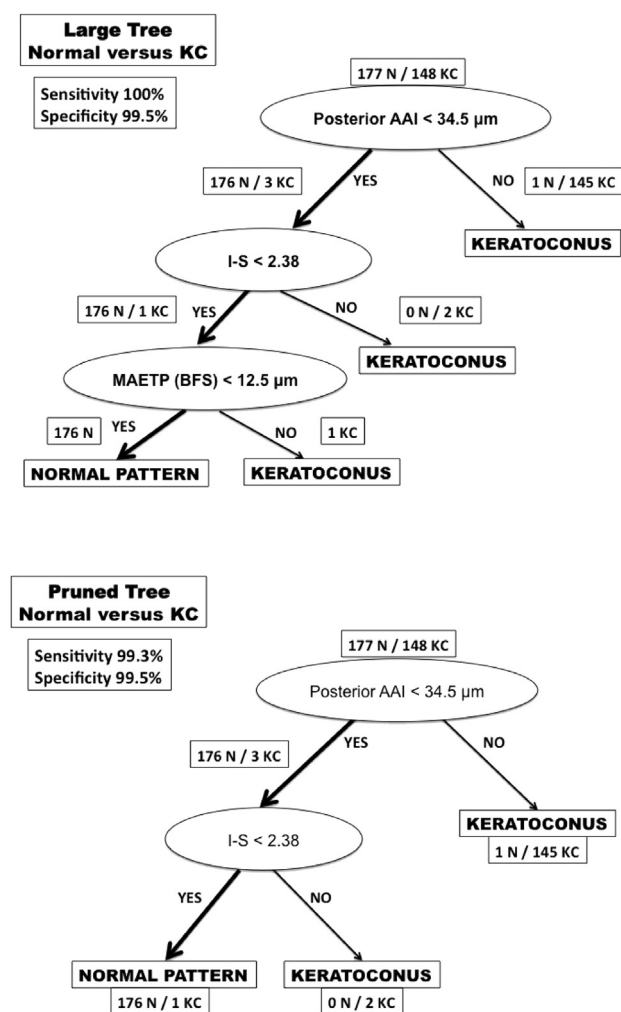


FIGURE 2. Decision trees generated using the automated decision tree classification for differentiating normal corneas (N) from keratoconus (KC); pairs of large and pruned trees. Bold arrows mean that eyes meet the condition expressed by the variable and thin arrows mean that eyes do not meet the condition expressed by the variable. (Top) Large tree. According to the discriminating rule, if an eye has a posterior asphericity asymmetry index (AAI) lower than 34.5 μm and if the inferior-superior (I-S) value is lower than 2.38, and if the maximum anterior elevation at the thinnest point (MAETP) in the best-fit sphere (BFS) display is less than 12.5 μm, it will be labeled as a normal cornea. This classification rule was tested with 100% sensitivity and 99.5% specificity. (Bottom) Pruned tree. The pruned tree discriminating rule includes only the first 2 variables and achieved 99.3% sensitivity and 99.5% specificity. Details of the splitting between classes of eyes are specified in the boxes at each node.

condition, such as keratoconus suspect, preclinical keratoconus, or forme fruste KC.⁶ However, although the strict criteria that define the KC suspect are not well established, it is generally accepted that forme fruste KC corresponds to the topographically normal eye that has a frank KC in the fellow eye,^{4,6,14} this definition being underpinned

by 2 major elements that characterize the KC. First, keratoconus is an asymmetric progressive disease where both eyes may have the same genetic makeup, and therefore, even the less-affected eye is thought to have a milder form of KC. In addition, longitudinal studies on unilateral KC have reported between 20% and 50% progression in clinically normal fellow eyes of patients with a unilateral KC after long follow-up periods, with a greater risk during the first 6 years of onset.^{7,15} Therefore, considering that these eyes should have the earliest topographic and tomographic manifestations of KC, it seemed appropriate to investigate these particular corneas for developing a screening tool for detecting subclinical forms of KC.

In the present study, the use of a machine learning classifier, the classification and regression tree, allowed for the differentiation between normal corneas and forme fruste KC with 93.7% sensitivity and 97.2% specificity with the large tree discriminating rule and 90% sensitivity and 86% specificity with the pruned tree discriminating rule (Figure 3). However, when developing a screening tool for detecting forme fruste KC, special attention should be paid to the generalizability of the discriminating rule found with the studied population. Large trees risk overfitting the training data, and therefore the discriminating rule generated would more likely reflect the characteristics of the different eye conditions included in our sample rather than being generalizable as is. In contrast, the pruning process applied to generate the pruned trees allows for improving the predictive accuracy of the decision tree and reducing the risk of error when such decision tree is applied to new samples.¹⁶ Therefore, by doing so, and pending further testing with new samples, pruned trees provide a better global solution that will generalize better than the locally optimal solution, but at the expense of sensitivity and specificity for the sample data. This distinction between large- and pruned-tree models explains the smaller number of variables used in the pruned tree and the slightly lower performance for discriminating normal from forme fruste KC, but at the same time it makes the 2 selected variables particularly meaningful for the clinician.

Although there is controversy about the first detectable sign of the ectatic disease—whether it would be first an inferior steepening seen in the anterior surface with Placido topography or subtle modifications of the posterior cornea detected by tomography—the discriminating rule generated for differentiating normal eyes from forme fruste KC has automatically selected variables that are related to the posterior corneal elevation and the corneal volume distribution. The posterior asphericity asymmetry index and the corneal volume were selected by the algorithm as the 2 most discriminant variables among the 55 parameters incorporated in the analysis for differentiating between normal eyes and forme fruste KC. The posterior asphericity asymmetry index, which is a quantitative indicator of the posterior surface asymmetry, was selected by the algorithm as the most discriminant parameter in both discriminating

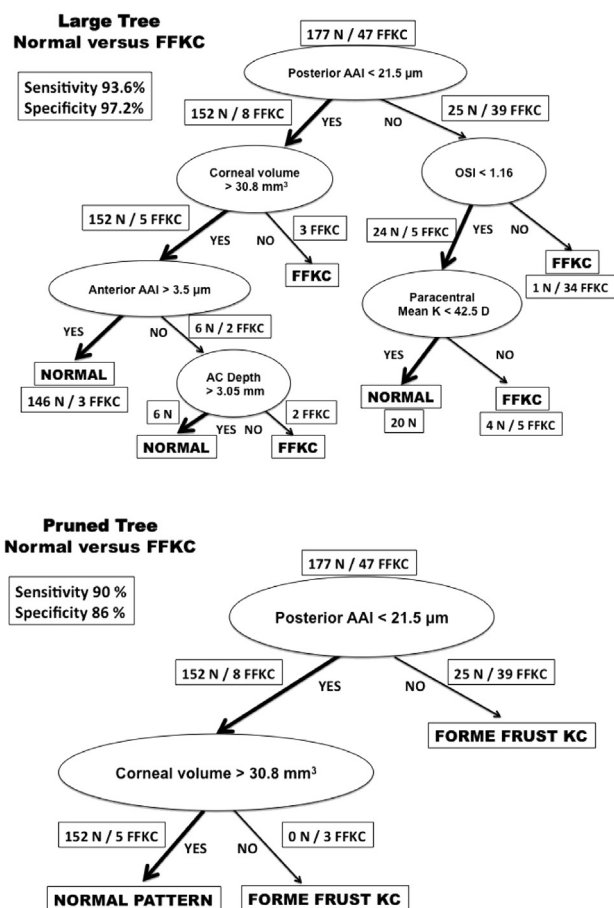


FIGURE 3. Decision trees generated using the automated decision tree classification for differentiating normal corneas from forme fruste keratoconus; pairs of large and pruned trees. The bold arrows mean that eyes meet the condition expressed by the variable and thin arrows mean that eyes do not meet the condition expressed by the variable. (Top) Large tree. The large tree discriminating rule includes 6 variables and achieved 93.6% sensitivity and 97.2% specificity. (Bottom) Pruned tree. According to the pruned tree discriminating rule, if an eye has a posterior asphericity asymmetry index greater than 21.5 μm and/or a corneal volume less than 30.8 mm^3 , it will be labeled as a forme fruste KC. This discriminating rule includes only 2 variables and was tested with 90% sensitivity and 86% specificity. Details of the splitting between classes of eyes are specified in the boxes at each node. N = normal eyes; FFKC = forme fruste of keratoconus; KC = keratoconus; LT = large tree; PT = pruned tree; AAI = asphericity asymmetry index; BFTA = best-fit toric and aspheric reference body; BFS = best-fit sphere reference body; TP = thinnest point; Kmax = keratometry maximum; AC = anterior chamber.

rules, normal eyes compared to KC (Pair 1) and normal eyes compared to forme fruste KC (Pair 2), with 2 different cutoff values, 34.5 μm and 21.5 μm , respectively. In a previous work, we demonstrated that the best-fit toric and aspheric reference surface, which is used for calculating

the asphericity asymmetry index, improved the sensitivity of the forme fruste KC detection when compared to the best-fit sphere surface.¹⁷ Indeed, by matching closer to the natural toric and aspherical shape of the cornea, the best-fit toric and aspheric body neutralizes the ridge pattern commonly seen in elevation maps calculated relative to a best-fit sphere and attributable to the effect of corneal toricity,^{18,19} and therefore would help to bring out the first signs of asymmetry in elevation. In the present study, the posterior asphericity asymmetry index was found statistically significantly higher in the forme fruste KC group than in the control group ($P = .002$), with a mean of $31.1 \pm 12.3 \mu\text{m}$ and $16.76 \pm 5 \mu\text{m}$, respectively. Corneal volume with a cutoff value of 30.83 mm^3 was selected by the algorithm for the second split of the decision tree (Pair 2). The relevance of the corneal volume distribution for differentiating normal cornea, KC, and subclinical KC has already been reported in the literature and the corneal volume has been reported to be lower in keratoconic corneas²⁰ and subclinical KC²¹ than in normal eyes. Similar results were found in our study, with a corneal volume significantly lower in forme fruste KC than in normal eyes ($P < .001$), with $31.9 \pm 1.7 \text{ mm}^3$ and $33.6 \pm 1.5 \text{ mm}^3$, respectively.

According to our findings, variables related to the anterior corneal surface were not selected in the upper part of the decision tree for discriminating normal eyes from forme fruste KC, which is not surprising given the definition of forme fruste KC included in our study, which involves normal topographic aspect with no asymmetry bowtie and no focal or inferior steepening pattern. Moreover, these results are consistent with many recent reports in the literature that consider the modifications of the posterior corneal surface and the corneal thickness distribution as key parameters in detecting the mildest forms of KC.^{4,22,23} Schlegel and associates,²² Pinero and associates,²¹ and De Sanctis and associates²⁴ have reported significant differences at the level of the posterior surface between normal eyes and subclinical KC, respectively, with the Orbscan system and with the Pentacam for the last 2. Saad and Gatineau⁴ have also highlighted the relevance of parameters that derived from the posterior surface as well as from the thickness spatial profile for identifying forme fruste KC. More recently, Arbelaez and associates have further supported this finding by demonstrating that the use of the posterior surface and thickness parameters significantly improved the sensitivity of subclinical KC detection among a large sample of patients.⁵ Our results are therefore in perfect agreement with these recent findings that corroborate the hypothesis that the earliest manifestations of KC would be detectable, in corneal imaging, by a concomitant corneal thinning with subtle modifications of the posterior surface.

Along with the growing interest in detecting subclinical KC and the new technological advances in corneal imaging, allowing for a more complete analysis of the

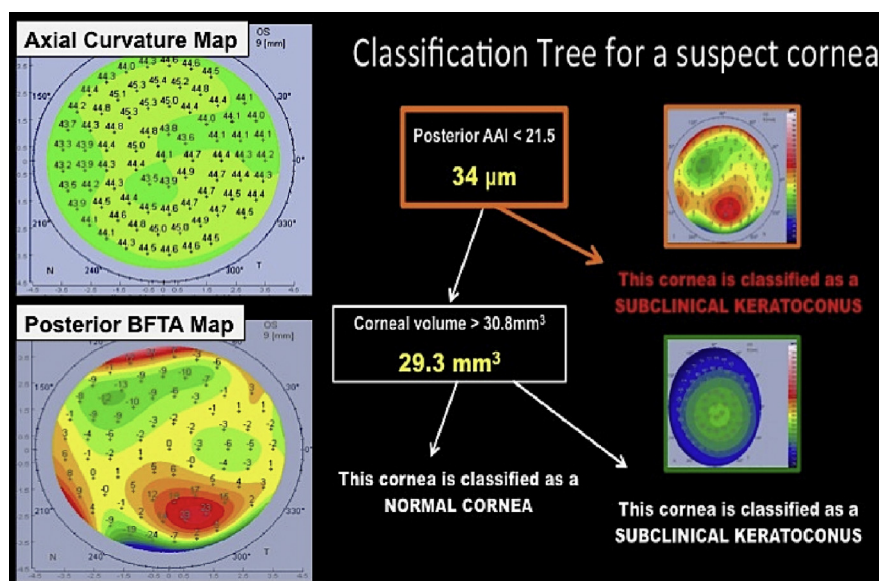


FIGURE 4. Example of forme fruste keratoconus differentiated from a normal cornea using the automated decision tree classifier. (Top left) The anterior curvature map expresses no asymmetry bowtie of focal inferior steepening, whereas (Bottom left) the posterior elevation map in best-fit toric and aspheric body display shows a clear inferior posterior bulging of 23 μm with a posterior asphericity asymmetry index calculated at 34 μm . Therefore, according to the discrimination rule built with this screening program (Right), the decision tree has identified this cornea as a subclinical KC. AAI = asphericity asymmetry index; BFTA = best-fit toric and aspheric reference body.

TABLE 2. Performances of the Discriminating Rules Generated Using an Automated Decision Tree Classification for Differentiating Normal Corneas From Forme Fruste Keratoconus and Normal Corneas From Keratoconus

	Sensitivity (%)		Specificity (%)	
	N vs FFKC	N vs KC	N vs FFKC	N vs KC
Number of eyes (subjects)	177 (95) vs 47 (47)	177 (95) vs 148 (102)	177 (95) vs 47 (47)	177 (95) vs 148 (102)
Large tree	93.6%	100%	97.2%	99.5%
Pruned tree	90%	99.3%	86%	99.5%

FFKC = forme fruste of keratoconus; KC = keratoconus; N = normal.

cornea, advanced screening tools for detecting subclinical KC have been developed with various imaging systems. The association of multiple corneal indices in 1 single discriminant function for detecting ectasia-susceptible cornea has been considered as an interesting approach for improving the sensitivity and specificity of such a screening tool²⁵ and has already been successfully tested with various systems.^{4,5,23}

The automated decision tree classification used in the present study is a data mining method that could be developed because of the availability of high-power computers and has the major advantage of being able to handle very large data sets, either large numbers of variables and/or large numbers of observations. The hierarchical discriminating rule produced is user friendly and can be displayed and easily understood, unlike the classical black box algorithm models found in the current imaging systems.

However, 1 of the limitations of this study is that the pruning process as well as the cross-validation method used in the present work, although helping to provide a better solution, still remain insufficient for generalizing the performance of the tree discriminating rule. They must be tested and validated in a new sample of unseen data before concluding on its generalizability. Indeed, the cross-validation model used here enabled validation of this screening program only for the subjects sampled in this study. Therefore, further evaluations with a larger sample of data will help to further validate and refine the performance of this test.

When comparing the performance of our screening program to others, Saad and Gatine¹⁴ reported the performance of a new score based on a discriminant function composed of multiple corneal indices for discriminating normal eyes from forme fruste KC with the Orbscan system

TABLE 3. Key Variables Selected in the Decision Trees by the Automated Decision Tree Classifier as the Most Discriminant Parameters for Differentiating Normal Corneas From Keratoconus and Normal Corneas From Forme Fruste Keratoconus: Means and Intergroup Comparisons

N (n subjects)	Means $\pm$ SD (Min; Max)			Intergroup Comparisons Mixed Model (P Value) ^a		
	Normal	FFKC	KC	Normal vs FFKC	Normal vs KC	FFKC vs KC
	177 (n = 95)	47 (n = 47)	148 (n = 102)			
Tree: normal vs FFKC						
Posterior AAI ^b	16.76 $\pm$ 5 (0; 36)	31.1 $\pm$ 12.3 (14; 66)	99.8 $\pm$ 49.2 (23; 250)	.002	< .001	< .001
Corneal volume	33.6 $\pm$ 1.5 (30.9; 38.5)	31.9 $\pm$ 1.7 (28.3; 35.9)	31.2 $\pm$ 1.7 (25.2; 35.4)	< .001	< .001	< .001
OSI	0.72 $\pm$ 0.4 (0.01; 1.8)	1.85 $\pm$ 1.2 (0.12; 5.3)	7.8 $\pm$ 3.9 (0.5; 21.3)	.002	< .001	< .001
Anterior AAI	9.2 $\pm$ 3.6 (0; 27)	15.8 $\pm$ 6.9 (3; 33)	54.4 $\pm$ 26.2 (13; 150)	.01	< .001	< .001
AC depth	3.2 $\pm$ 0.3 (2.4; 3.8)	3.2 $\pm$ 0.3 (2.4; 3.8)	3.3 $\pm$ 0.3 (2.6; 3.9)	.98	.07	< .001
Tang MeanK (4-7 mm)	41.6 $\pm$ 1.4 (38.4; 44.7)	41.2 $\pm$ 1.8 (37.6; 44.8)	40.6 $\pm$ 2.7 (33.8; 48)	.23	< .001	< .001
Tree: Normal vs KC						
Kmax (D)	44.2 $\pm$ 1.3 (41.2; 46.9)	45.4 $\pm$ 1.7 (41.7; 49.4)	54.7 $\pm$ 5.2 (47; 76.6)	.01	< .001	< .001
I-S	0.58 $\pm$ 0.4 (0; 1.95)	0.96 $\pm$ 0.5 (0.1; 2.3)	8.44 $\pm$ 4.3 (0.9; 24.7)	.004	< .001	< .001
BFS MAETP	1.55 $\pm$ 1.4 (-2; 8)	3.3 $\pm$ 2.5 (-1; 11)	17.9 $\pm$ 10.8 (-2; 56)	.04	< .001	< .001

AC = anterior chamber; AAI = asphericity asymmetry index; BFS = best fit sphere; BFTA = best-fit toric aspheric surface; FFKC = forme fruste keratoconus; I-S = inferior-superior value; KC = keratoconus; Kmax (D) = maximal keratometry; MAETP = maximum anterior elevation at the thinnest point; MeanK = average keratometry (diopters); OSI = opposite sector index; SD = standard deviation; Tang = tangential map.

^aBold = statistically significant.

^bVariable used in both trees (Normal vs FFKC and Normal vs KC), but with a different cutoff value.

and found similar results to our large tree discriminating rule, with 93% sensitivity and 92% specificity, whereas they reached a slightly higher sensitivity than our pruned tree discriminating rule. However, the larger sample size analyzed in the current study and the difference in the number of predictors (34 in their study compared to only 2 in the pruned tree) used in the algorithm for discriminating normal eyes from 40 forme fruste KC might explain the differences in performance. Additionally, although successfully tested retrospectively on 1 case of post-LASIK ectasia,²⁶ the index newly developed by Saad and Gatinel still remains to be validated on new data sets. In contrast, the pruned tree discriminating rules are based on cross-validation before the final tree is selected and therefore provide a more generalizable performance than the locally optimal solution, although validation studies on different larger data sets should be performed to further validate the discriminating ability of our rule. More recently, Arbelaez and associates developed a new classification algorithm for the detection of subclinical KC with the Sirius system using a machine learning technique, the support vector machine.⁵ The authors reported excellent results on a very large sample with 92% sensitivity and 97.7% specificity for detecting subclinical KC. However, they included in the subclinical KC group corneas that already had topographic pattern of localized steepening but no clinical signs of KC, which might explain in part the difference between our respective findings. Comparison with the Belin-Ambrosio Enhanced Ectasia display of the Pentacam system (Oculus Optikgeräte GmbH, Wetzlar, Germany) is not possible, since it has not been tested yet

and the models based on multiple regression analysis proposed by Uçakhan and associates²³ with several Pentacam indices are not available in the current software of the Pentacam. However, the discriminating ability of their model was lower than what we found in the present study, with 77.3% and 92.1% in their sample, compared to 93.6% and 97.2% of sensitivity and specificity, respectively, in our sample, although the sample size was smaller in their study and the selection of their subclinical KC population was different and based on suspect topographic patterns.

Decision tree classifications have already been used in previous studies but never in an attempt to facilitate the detection of subclinical KC. Maeda and associates have first proposed the use of such an approach for classifying abnormal topographic patterns.²⁷ More recently, Twa and associates used the automated decision tree classification approach for discriminating between KC and normal corneal shape and achieved 92% sensitivity and 93% specificity.²⁸ In addition, the performance of the decision tree classifier was compared, in terms of discriminating ability, to other KC detection algorithms, such as the KISA% index, the Keratoconus Prediction Index, the modified Rabinowitz-McDonnell index, the Cone Location and Magnitude Index, and the Z3, third-order Zernike Polynomial index, and has been found better than 4 of the 5 other detection methods, whereas it performed equally with the Z3 index.

In the present study, the use of an automated decision tree classifier allowed the detection of KC and forme fruste KC with high sensitivity and a similar discriminating ability to what has been previously reported with different imaging systems. In addition, this approach enabled us to

reveal the clinical significance and relevance of certain parameters, such as the posterior asphericity asymmetry index and the corneal volume. Although it remains to be

further validated in a new larger sample, such an artificial intelligence system might undoubtedly help the clinician in detecting mild ectatic corneas before refractive surgery.

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Biosketch

David Smadja, MD, was born and raised in France, where he completed his Medical School and Residence Program in the Department of Ophthalmology at Bordeaux Hospital University. Subsequently, he completed a Research Fellowship in Cornea and Refractive Surgery at the Cole Eye Institute, at the Cleveland Clinic Foundation with Ronald R. Krueger from 2010 to 2011. After completion of his research fellowship, he graduated from the European Board of Ophthalmology and completed a Clinical Fellowship in Anterior Segment and Refractive Surgery at the Bordeaux University Hospital under the supervision of Professor Joseph Colin. Currently, Dr Smadja is involved in many scientific projects in refractive surgery and in the National Reference Center for Keratoconus.